



OFFICIAL CITIZEN & OPERATOR DOCUMENTATION

USER OPERATION MANUAL

صارفین اور شہریوں کیلئے معلوماتی رہنما

Comprehensive guide for public recycling, interactive screen navigation, multi-sensor deposit validation, points redemption, and intelligent kiosk display features.

**7-Step Flow**

Frictionless deposit from Key 0 trigger to wallet credit and live rating.

**Idle Expansion**

50% screen instructional expansion with active digital signage playback.

**Instant Rewards**

Real-time mobile wallet sync with Central Cloud and SQL Server ledger.

DOCUMENT ID

DOC-RVM-UM-2026-V3.2

DATE / RELEASE

September 2026

APPLICABILITY

RVM Kiosk Model RVM-001

CLASSIFICATION

Public / User Manual

Manual Overview & Navigation

A structured roadmap of all user interactions, kiosk screen displays, and environmental reward steps.

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⚡ QUICK START FOR CITIZENS To recycle: Walk up to the RVM kiosk → Press '0' on the keypad → Insert your empty bottle/can bottom-first → Wait for the green checkmark → Press Enter → Enter your 11-digit mobile number → Rate your experience → Collect your reward points!	

1. Reverse Vending Machine Overview

An intelligent, multi-material automated recovery unit engineered for university campuses and public plazas.

DUAL-DISPLAY KIOSK DESIGN

The RVM Kiosk employs a dual-display architecture engineered for maximum citizen engagement and continuous environmental education:

- Interactive Portrait Kiosk Display (Screen 1 - 1080x1920):** The primary operational interface where citizens view animated instructions, live telemetry, sensor sizing results, point balances, and leaderboard standings.
- Digital Signage Video Screen (Screen 0 / Lower Display):** Continuously loops high-definition environmental public service announcements (PSAs), commercial partner advertisements, and sustainability metrics without disrupting the active recycling transaction.



SUSTAINABILITY MISSION

Every container deposited into the RVM directly prevents municipal landfill overflow and saves precious natural resources: **+0.15 kg CO2** emission prevented and **+0.75 Liters** of clean water conserved per verified container.

KEY CITIZEN TOUCHPOINTS

Touchpoint Component	Citizen Function
Illuminated Aperture	Circular entry slot with white LED guide ring. Accepts PET bottles, cans, and UBC cartons.
Motorized Safety Gate	Anti-pinch protective door opens only when the citizen indicates readiness by pressing 0.
Physical Matrix Keypad	12-key tactile keypad mounted at ergonomic height. Resistant to rain, dust, and direct sunlight.
Instructional Container	High-visibility animated display showing real-time feedback, sensor diagnostics, and celebration animations.
Reward Balance Display	Instant point tracker displaying accumulated session points in 56pt high-contrast bold digits.

CITIZEN SAFETY NOTICE

The intake chamber is monitored by safety sensors. Never attempt to force foreign objects, glass, or pressurized cans into the machine. The safety door will immediately halt if an obstruction is detected.

SUPPORTED RECYCLABLE MATERIALS



Plastic PET Bottles

Water, soda, juice bottles (250ml to 2.25L). Must be empty of liquid. S, M, L accepted.



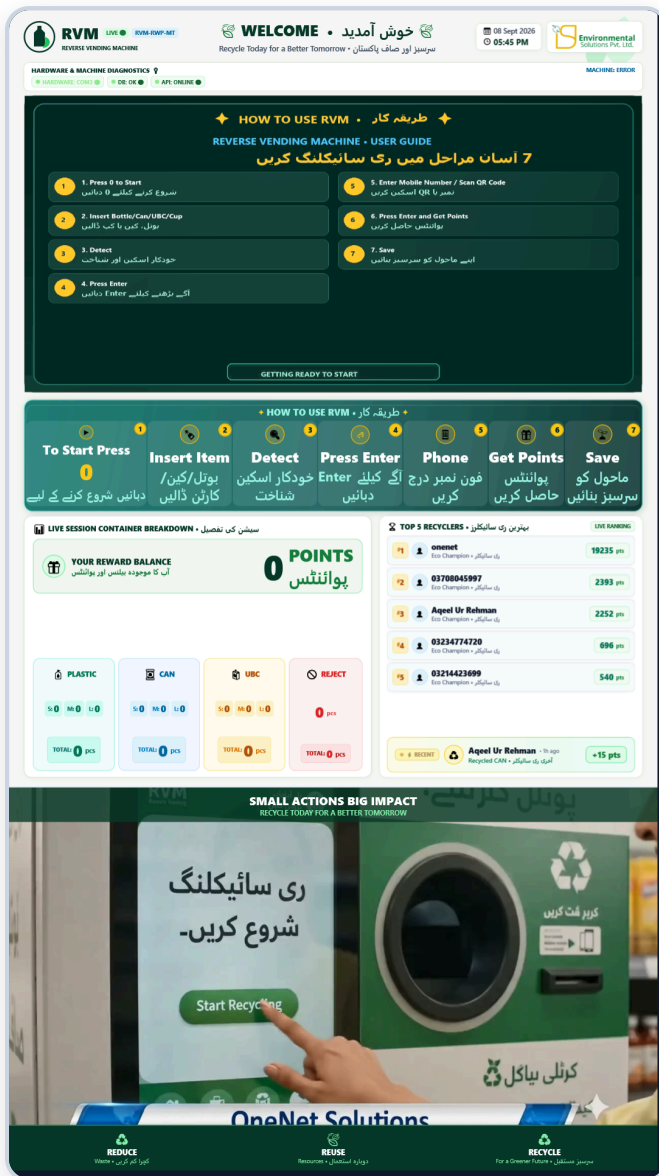
Aluminium Cans

Beverage cans, soda tins. Verified by internal inductive sensor. S, M, L accepted.



UBC TetraPak Cartons

Juice cartons, milk packs. Verified by ultrasonic acoustic bounce. S, M, L accepted.



STANDBY SCREEN · SCREEN 02

Home Page & Citizen Dashboard

مرکزی اسکرین اور طریقہ کار

When the RVM is in its ready standby state, it presents the primary welcome screen. The interface is meticulously balanced with high-contrast typography, emerald eco-card containers, and bilingual Urdu and English text.

HOW TO BEGIN

Press the '0' key on the keypad or tap the glowing green button on the screen.

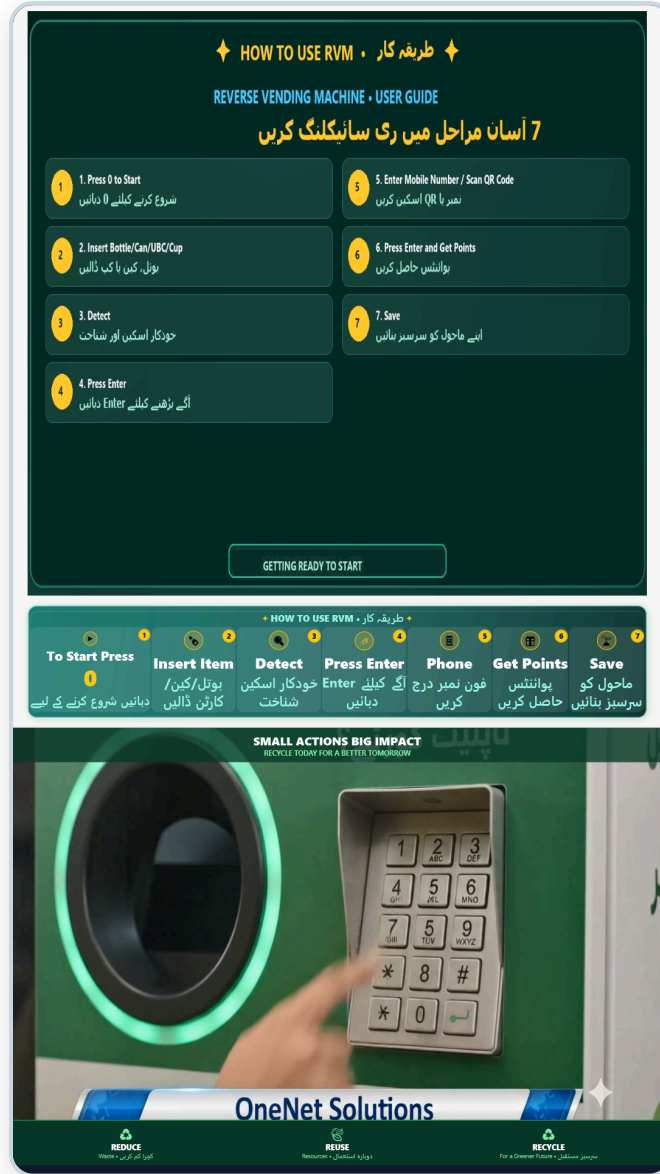
Dashboard Section	Citizen Feature & Information
Top Header Bar	Machine ID (RVM-001), Real-time Live Status badge, and current Pakistan Standard Time.
Hardware Diagnostics	Green status dots confirming Serial connection, Local Database (OK), and Central Cloud API (ONLINE).
Instructional Player	16:9 high-definition video widget running bilingual instructional motion guides.
How to Use RVM Bar	Teal glassmorphism card outlining the 7 chronological steps of the recycling journey.
Session Breakdown	Live material tallies (Plastic, Cans, UBC, Rejected) and large 56pt Reward Balance.
Top 5 Eco Champions	Live community leaderboard celebrating top campus recyclers and recent recycling activity.

COMMUNITY RECOGNITION

Top campus recyclers are automatically ranked on the right-hand leaderboard. Your name and reward points will be showcased to motivate the community!

2.2 Kiosk Idle Mode & Display Expansion

Automatic display optimization when the kiosk is inactive for more than 1 minute to attract pedestrians and showcase educational content.



STANDBY MODE - 60s INACTIVITY

1-Minute Inactivity Idle Screen

ایک منٹ بعد اسکرین کی خودکار توسیع

When no citizen interacts with the kiosk for over **1 minute (60 seconds)**, the application automatically transitions into cinematic standby mode:

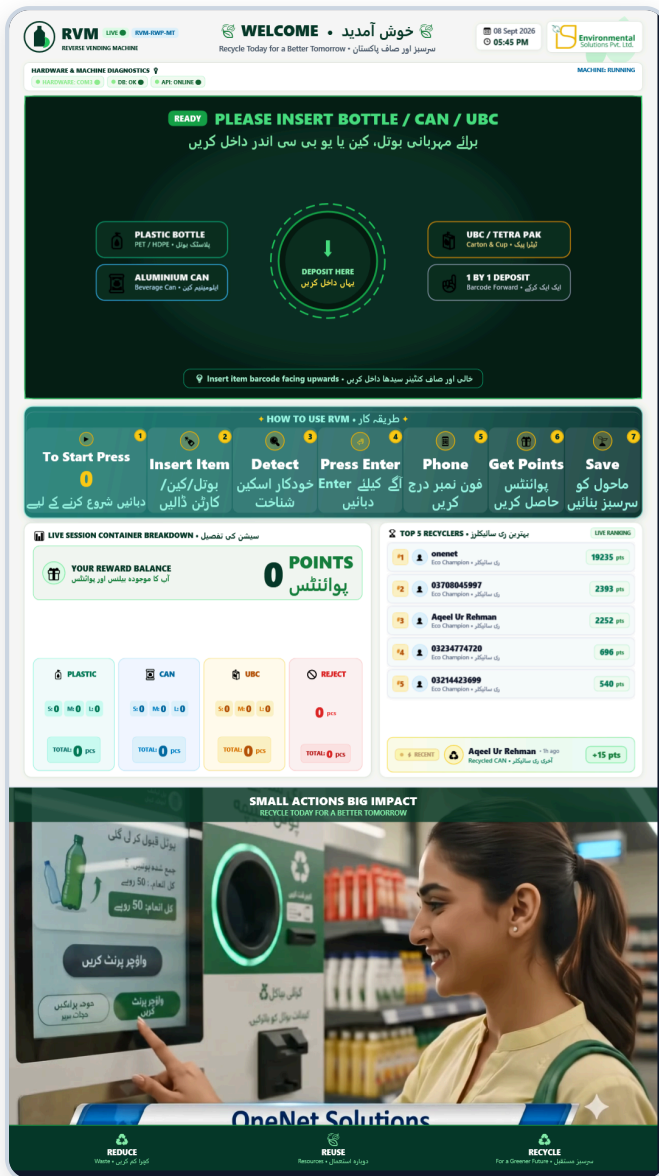
Layout Element	Behavior During Idle Mode
Instructional Video	Expands to 50% of screen height (960px) with smooth zoom-in styling inside container.
How to Use RVM Card	Stays 100% visible with all 7 step cards and Urdu text clearly legible below the video.
Ad Video Section	Stays 100% visible & playing in the remaining lower 740px screen area.
Header & Status Bar	Automatically fade out and collapse to 0 height for clean visual immersion.
Lower Dashboard	Live session breakdown & leaderboard collapse to 0 height.
Zero Overlap Geometry	Strict Auto row stacking guarantees zero visual clipping or touching between cards.

HOW TO RETURN TO DEFAULT SCREEN

Press the '0' key on the keypad or touch any area of the screen. The kiosk instantly zooms out and restores the default screen!

⚡ TESTING HOTKEY

Technicians and attendants can instantly toggle Idle Mode anytime by pressing **Ctrl + I** without waiting for the full 60 seconds.



STEP 1 OF 7 • KEYPAD KEY '0'

Starting Your Recycling Session

پہلا مرحلہ: سیشن کا آغاز اور تیاری

To start a recycling session, press the '0' key on the physical hardware keypad or touch the circular start button. The RVM acknowledges citizen presence immediately: the motorized entrance gate opens, internal chamber lights turn on, and sensors arm for container entry.

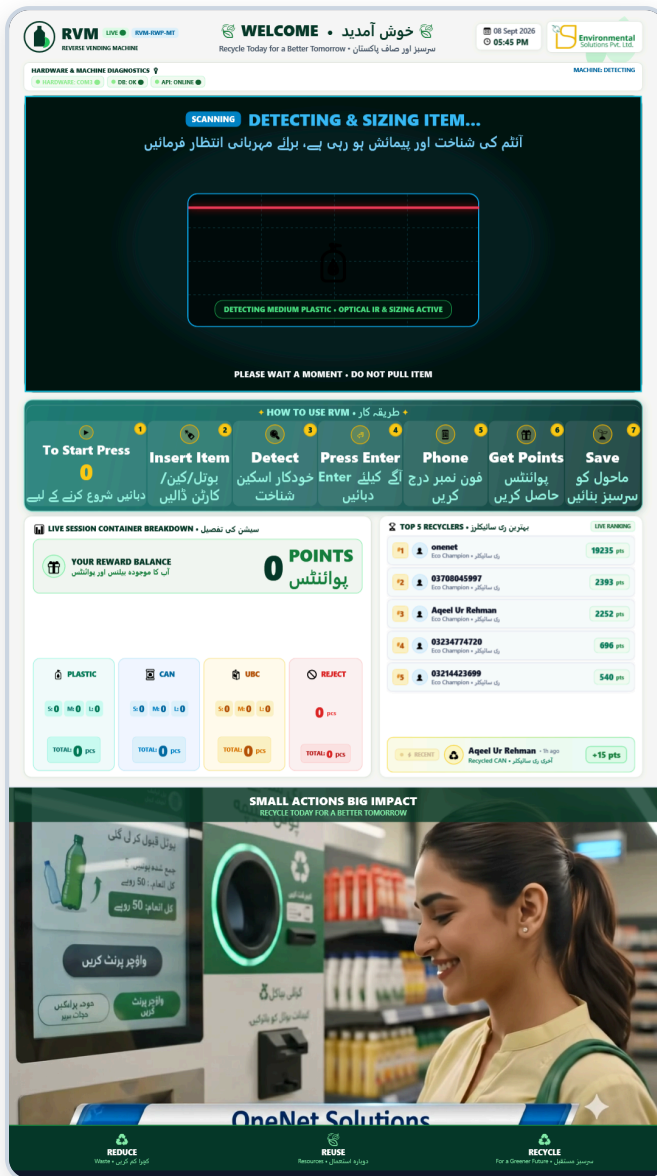
CITIZEN ACTION REQUIRED

Ensure your bottle or can is completely empty. Insert container bottom-first into the illuminated circular aperture.

Mechanical & Electronic Event	Technical State
Entrance Door Servo	Rotates 90° to open position (Aperture Ready)
Chamber Lighting	High-intensity white LED ring illuminates interior
Acoustic Sensor Ping	Ultrasonic transducer arms (Pin D12/D13)
Safety Gate Sensor	Infrared safety beam active to prevent finger pinching
Session Timer	30-second activity countdown initiated

TIP FOR OPTIMAL RECYCLING

Always insert bottles bottom-first with the cap facing toward you. Keep barcodes visible if present to assist high-speed classification.



STEP 2 OF 7 • DETECTION & SIZING

Real-Time Sensor Scanning

دوسرا مرحلہ: اسکیننگ اور پیمائش

As the container settles inside the chamber, the RVM executes a sub-second multi-sensor sweep. The system cross-references optical height beams, inductive metallic response, and ultrasonic distance to classify material and calculate dimensions.

CITIZEN ACTION REQUIRED

Allow container to rest inside chamber. Wait 1 second while sensors verify dimensions and confirm acceptance.

Sensor Subsystem	Detection Criteria & Result
Inductive Sensor (D5)	Metal conductivity check (1 = Can / Metal, 0 = Plastic / UBC)
IR Optical Array (D2, D7, D8)	3-tier height measurement: Small (<16cm), Medium (16-24cm), Large (>24cm)
Ultrasonic Chamber (D12/D13)	Precise length measurement via high-frequency acoustic bounce
Liquid Detection	Weight & optical refraction test to detect residual liquids
Arbitration Time	Under 850 milliseconds for complete item classification

ACCEPTED SIZE CATEGORIES

Small: 250ml–350ml (+5 pts) • **Medium:** 500ml–1.0L (+10 pts) • **Large:** 1.5L–2.25L (+15 pts). All sizes are automatically awarded appropriate points!

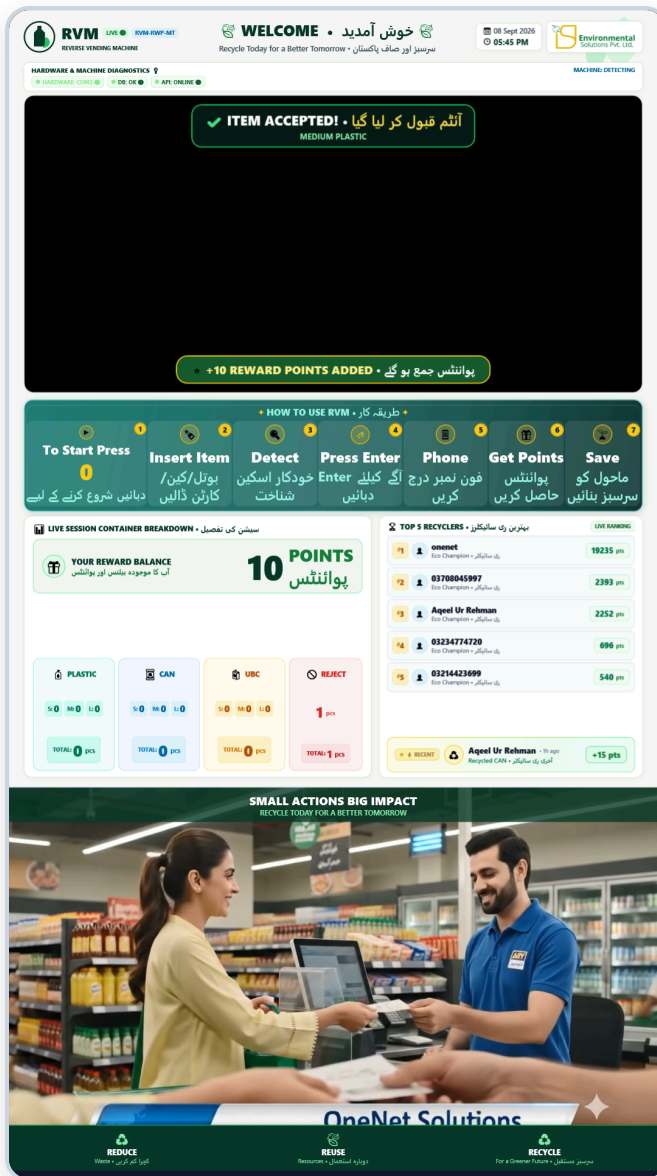
Interactive Motion Overlays on Video Widget

During the recycling transaction, the video player displays rich dynamic overlays that guide the citizen through detection stages.

STATE 1 Please Insert Item PLEASE INSERT BOTTLE / CAN برائے مہربانی خالی بوتل یا کین ڈالیں Displays an animated pulsating circular aperture showing bottles entering smoothly. Reassures citizens that the machine is armed and awaiting container deposit.	STATE 2 Detecting & Sizing DETECTING & SIZING ITEM... بوتل کی شناخت اور پیمائش ہو رہی ہے Features a simulated bidirectional laser beam scanning vertically across the container. Confirms to the citizen that sensors are measuring material dimensions in real-time.
STATE 3 Accepted + Points ITEM ACCEPTED! +10 PTS بوتل قبول! پوائنٹس کا اندراج ہو گیا Celebratory green notification with confetti graphics. Automatically plays material dancing animation (Plastic, Can, or TetraPak) matching the recognized item.	STATE 4 Item Rejected ITEM REJECTED! PLEASE RETRIEVE آئٹم نامنظور! برائے مہربانی واپس نکال لیں Clear safety warning with gentle audio alert. Gate unlocks to allow retrieval of non-recyclable garbage, glass, or containers with remaining liquid.

The screenshot displays the RVM Kiosk interface during a rejection state. At the top, a green banner reads "HOW TO USE RVM • طریقہ کار •". Below this, a row of seven numbered steps is shown: 1. To Start Press (دبائیں شروع کرنے کے لیے), 2. Insert Item (بوتل/کین/کارٹن ڈالیں), 3. Detect (خودکار اسکین شناخت), 4. Press Enter (اگے کیلئے Enter دبائیں), 5. Phone (فون نمبر درج کریں), 6. Get Points (پوائنٹس حاصل کریں), and 7. Save (ماحول کو سرسبز بنائیں). The main area is divided into two sections. The left section, titled "LIVE SESSION CONTAINER BREAKDOWN • سیشن کی تفصیل", shows the "YOUR REWARD BALANCE" as 0 POINTS (پوائنٹس). Below this, four container categories are listed: PLASTIC (TOTAL: 0 pcs), CAN (TOTAL: 0 pcs), UBC (TOTAL: 0 pcs), and REJECT (TOTAL: 1 pcs). The right section, titled "TOP 5 RECYCLERS • بہترین ری سائیکلرز", shows a "LIVE RANKING" list of five users with their scores: 1. onenet (19235 pts), 2. 03708045997 (2393 pts), 3. Aqeel Ur Rehman (2252 pts), 4. 03234774720 (696 pts), and 5. 03214423699 (540 pts). At the bottom right, a "RECENT" section shows a notification for "Aqeel Ur Rehman" who recycled a CAN, earning +15 pts.

Figure 7.1: Real Kiosk Rejection Screen displaying safety alert and retrieval instructions.



STEP 4 OF 7 • ANTI-CHEAT VALIDATION

Drop Confirmation & Claiming Rewards

چوتھا مرحلہ: بوتل چیمبر میں گرنے کی تصدیق

To maintain reward integrity, points are only awarded after the anti-cheat drop gate actuates 180° and the ultrasonic sensor verifies the physical fall of the container into the internal bin. Once confirmed, the citizen can insert another item or press **Enter** to proceed to wallet credit.

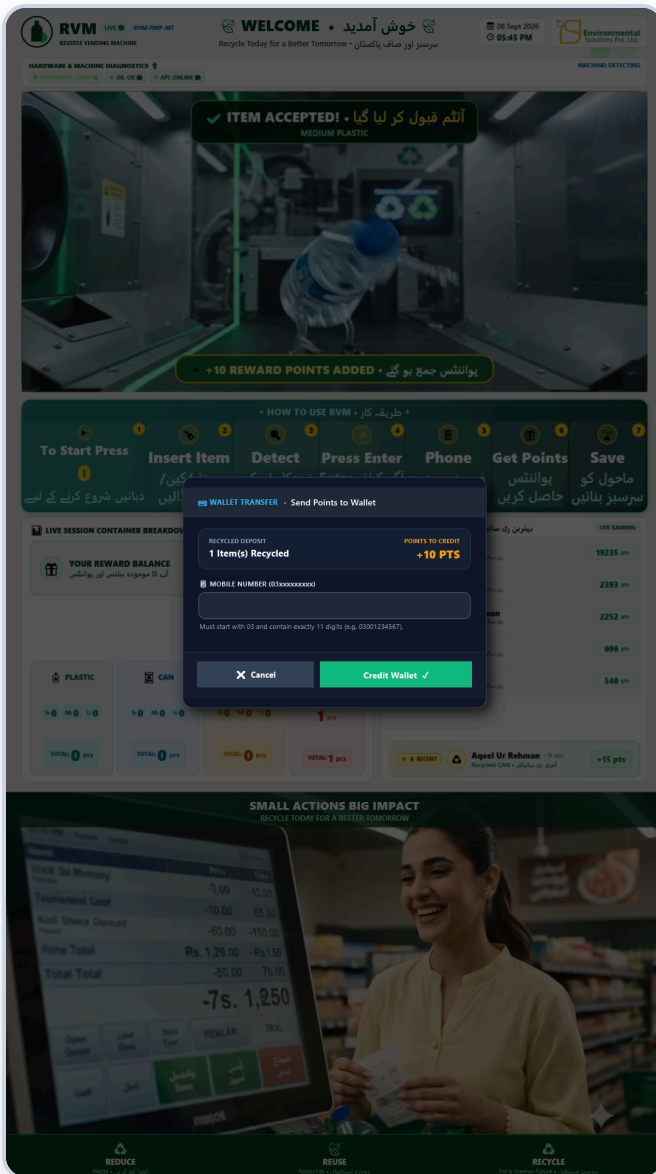
CITIZEN ACTION REQUIRED

Insert another container to earn more points, OR press 'Enter' on the keypad to finalize session and claim rewards.

Verification Parameter	Anti-Cheat Mechanism
Drop Gate Actuator	Servo Pin D8 rotates 180° downward to drop container into bin
Ultrasonic Fall Ping	Acoustic bounce detects item clearing chamber (BOTTLE:CLEARED)
String-Tie Anti-Cheat	If an item is pulled back via string, machine flags ERROR:CLEAR_TIMEOUT
Session Accumulation	Points and items increment live on dashboard breakdown

🎁 MULTI-CONTAINER SESSIONS

You do not need to enter your phone number for each bottle! Keep depositing containers one after another. When you are completely done, press **Enter** once to credit all accumulated points simultaneously.

**STEPS 5 & 6 • MOBILE IDENTIFICATION**

Entering Mobile Number & Syncing Rewards

پانچواں مرحلہ: انعام کیلئے موبائل نمبر کا اندراج

After pressing Enter, the modal prompts the citizen for their 11-digit mobile phone number (03xxxxxxx). This number acts as your unique recycling account, aggregating points across all campus and public RVM machines.

CITIZEN ACTION REQUIRED

Type your 11-digit mobile number using the keypad digits (0-9). Press 'Enter' when finished to submit.

Input Validation Rule	Specification
Mobile Number Format	Standard Pakistani format: 03001234567 (Strict 11 Digits)
Backspace Correction	Pressing '*' key clears the last typed digit
Cancel Transaction	Pressing '#' key cancels entry and returns to home
Cloud Synchronization	Instant HTTPS API commit to Central Cloud database
Local Database Commit	Instant transaction commit to local SQL Server ledger

POINTS REDEMPTION

Your points never expire! Accumulated points can be converted into mobile balance top-ups, cafeteria discount vouchers, and campus bookstore gift coupons.

STEP 7 OF 7 • CITIZEN EXPERIENCE RATING

Rating Your Recycling Experience

ساتواں مرحلہ: تجربے کی درجہ بندی اور رائے

Following point crediting, the kiosk presents a 1-to-5 star experience rating screen. Citizens can touch the stars to submit their rating and tap quick feedback tag chips ("Fast & Easy", "Clean Machine", "Great Rewards").

CITIZEN ACTION REQUIRED

Tap 1 to 5 stars on the screen or press a keypad digit (1-5), then press Enter to submit your feedback.

Feedback Parameter	Citizen Feature
Star Rating Scale	1 to 5 Gold Stars (Poor to Excellent)
Feedback Tags	Fast & Easy, Clean Machine, Great Rewards, Needs Help
Auto-Skip Timeout	10-second timer automatically completes if citizen walks away
Final Receipt Screen	Displays total points earned and CO2 prevented
Auto-Reset	Machine cleanly resets session buffer for next user

☀️ CONTINUOUS IMPROVEMENT

Your feedback is transmitted directly to the University Administration and Municipal Waste Management dashboard to ensure the kiosk remains clean and well-maintained.

4. Keypad & Input Quick Reference

Comprehensive guide to all tactile keypad inputs, hotkeys, and citizen commands.

Key / Input	Screen State	Function & Kiosk Response
Key 0	Standby / Idle Mode	Hardware Session Trigger: Opens safety door, activates chamber illumination, and arms sensors. Also immediately exits Idle Expanded Mode and restores default screen.
Enter ↵	Container Deposited	Finalize & Proceed: Concludes container intake and triggers the mobile wallet number entry modal.
Enter ↵	Mobile Number Modal	Submit Wallet: Validates 11-digit mobile number, credits points to database, and navigates to rating screen.
Digits 0–9	Mobile Number Modal	Phone Number Input: Types digits for Pakistani mobile number (03xxxxxxxx).
Star Key (*)	Mobile Number Modal	Backspace / Delete: Removes the last typed digit for instant correction.
Hash Key (#)	Any Modal	Cancel / Back: Closes active modal dialog and returns to the previous screen.
Touch Screen	Any Screen	Direct Touch: Citizens can tap on-screen buttons, star ratings, and feedback chips directly.
Code 001	Standby Home Screen	Direct Demo Mode: Typing 0-0-1 on keyboard starts the live demo session directly on the kiosk screen without dialogs.
Ctrl + I	Any State	Idle Mode Test Hotkey: Instantly toggles the 50% expanded instructional video standby mode.



KEY 0 HARDWARE RESERVATION

Key 0 is strictly reserved for physical hardware when connected. When Arduino COM port is online, pressing 0 initiates the physical gate and sensor loop. In disconnected testing mode, typing 001 initiates demo mode.

ERGONOMIC ACCESSIBILITY

The hardware keypad features embossed numbers and high-contrast tactile feedback compliant with universal accessibility standards for wheelchair users and visually impaired citizens.

5. Frequently Asked Questions & Troubleshooting

Common questions regarding recycling rewards, container acceptance, and citizen assistance.

Q1: Why did the machine reject my plastic bottle?

Answer: The most common cause is residual liquid inside the bottle. Optical sensors detect liquid refraction and reject the bottle to prevent sticky contamination inside the bin. Please empty all liquids completely before inserting. Rejection can also occur if the bottle is severely crushed or is an unsupported plastic type (e.g. PVC or HDPE detergent jugs).

Q2: What happens if the internet connection is temporarily offline?

Answer: Your points are 100% safe! The RVM kiosk operates a resilient local Microsoft SQL Server database. When the cloud connection is offline, points are saved locally in an offline queue and automatically synchronized to the Central Cloud server as soon as the network reconnects.

Q3: How many containers can I deposit in a single session?

Answer: There is no limit! You can insert as many bottles, cans, and cartons as you wish during one session. The live counter tracks every accepted container. When you are finished, press **Enter** to claim all points together.

Q4: How do I redeem my points for balance or gifts?

Answer: Visit the official web portal at isprvm.binishaqsoft.com or visit the campus partner desk. Enter your registered 11-digit mobile number to view balance vouchers, mobile recharge codes, and gift rewards.

CITIZEN SUPPORT & HELPLINE

If an item is stuck or you require assistance: Call Kiosk Support at **0800-RECYCLE (0800-73292)** or email support@binishaqsoft.com. State the Machine ID displayed in the top header (e.g. **RVM-001**).